Analysis Tickets:

Topological and Filter-Theoretic Version with Practice Problems

Metric Spaces, Compactness, Uniform Convergence, Integrals, and Power Series

A rewritten lecture handout using neighborhood filters, ultrafilters, Cauchy filters, metric uniformities, and exam-oriented practice problems.

This handout rewrites a standard first analysis course in a more structural language. The order of ideas is:

metric spaces $\Rightarrow$ topologies $\Rightarrow$ neighborhood filters $\Rightarrow$ compactness via ultrafilters $\Rightarrow$ uniformities and Cauchy filters.

When a theorem is originally metric, the abstract formulation is immediately expanded back into the usual ε-δ or sequence statement.

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Contents

How to use these notes

A correct oral answer in analysis should contain four parts: the exact definitions, the theorem with all hypotheses, the proof mechanism, and a counterexample or warning showing why the hypotheses matter. The language of filters is useful because it makes several different-looking statements identical:

$$\text{continuity} = \text{preservation of convergent filters,}$$

$$\text{compactness} = \text{convergence of ultrafilters,}$$

$$\text{completeness} = \text{convergence of Cauchy filters,}$$

$$\text{uniform continuity} = \text{preservation of entourages.}$$

The price is that one must distinguish carefully between topology and uniform structure. Ordinary continuity is topological; uniform continuity and Cauchy behavior are not purely topological.

Conventions

Unless stated otherwise, (X, d), (Y, ρ), and (Z, δ) are metric spaces. Their induced topologies are used silently. The scalar field $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. For a topological space X, $\mathcal{N}_X(x)$ denotes the neighborhood filter of x. For a map $f : X \to Y$, $f_*\mathcal{F}$ denotes the image filter.

Ticket 1. Metric spaces as topological spaces

Core idea. A metric gives numerical distances, but the first structure used by continuity is the topology generated by metric balls.

Definition 1.1 (Metric)**.** *A metric on a set X is a function $d : X \times X \to [0, \infty)$ such that, for all $x, y, z \in X$,*

$$d(x, y) = 0 \iff x = y, \qquad d(x, y) = d(y, x),$$
$$d(x, z) \le d(x, y) + d(y, z).$$

The pair (X, d) is a metric space.

Definition 1.2 (Metric topology)**.** *For $a \in X$ and $r > 0$, put*

$$B_d(a, r) = \{x \in X : d(x, a) < r\}.$$

A set $U \subset X$ is open if for every $x \in U$ there exists $r > 0$ such that $B_d(x, r) \subset U$. This defines the topology induced by d.

Proposition 1.1 (Topology axioms)**.** *The open sets induced by a metric satisfy:*

1. *$\varnothing$ and X are open;*
2. *arbitrary unions of open sets are open;*
3. *finite intersections of open sets are open.*

Proof. The first two statements are immediate. For finite intersections, let $x \in U_1 \cap \cdots \cap U_n$, where each U_i is open. Choose $r_i > 0$ such that $B(x, r_i) \subset U_i$. Then with $r = \min_i r_i$,

$$B(x, r) \subset U_1 \cap \cdots \cap U_n.$$

Thus the intersection is open. □

Definition 1.3 (Neighborhood filter)**.** *Let X be a topological space and $x \in X$. The neighborhood filter at x is*

$$\mathcal{N}_X(x) = \{A \subset X : \exists \text{ open } U \text{ with } x \in U \subset A\}.$$

In a metric space this is equivalently

$$\mathcal{N}_X(x) = \{A \subset X : \exists r > 0,\ B_d(x,r) \subset A\}.$$

Proposition 1.2. *For each $x \in X$, $\mathcal{N}_X(x)$ is a filter on X.*

Proof. The empty set is not a neighborhood of x. If $A, B \in \mathcal{N}_X(x)$, choose open sets U, V with $x \in U \subset A$ and $x \in V \subset B$. Then $U \cap V$ is open and

$$x \in U \cap V \subset A \cap B,$$

so $A \cap B \in \mathcal{N}_X(x)$. If $A \in \mathcal{N}_X(x)$ and $A \subset C \subset X$, then the same open neighborhood contained in A is contained in C, hence $C \in \mathcal{N}_X(x)$. □

Proposition 1.3 (Equivalent finite-dimensional metrics)**.** *On $\mathbb{R}^n$, the metrics*

$$d_1(x,y) = \sum_{k=1}^{n} |x_k - y_k|, \quad d_2(x,y) = \left(\sum_{k=1}^{n} |x_k - y_k|^2\right)^{1/2}, \quad d_\infty(x,y) = \max_k |x_k - y_k|$$

generate the same topology and therefore the same neighborhood filters.

Proof. For all $x, y \in \mathbb{R}^n$,

$$d_\infty(x,y) \le d_2(x,y) \le d_1(x,y) \le n d_\infty(x,y).$$

Hence every ball for one metric contains a smaller ball for the others. Therefore the open sets and the neighborhood filters coincide. □

Common mistake. Same topology does not always mean same uniform structure. For example, on $\mathbb{R}$, the usual metric $d(x,y) = |x-y|$ and $\rho(x,y) = |\arctan x - \arctan y|$ generate the same topology, but the sequence n is ρ-Cauchy and not d-Cauchy.

Oral-exam minimum.

- Define the metric topology through balls.
- Define the neighborhood filter $\mathcal{N}_X(x)$.
- Prove that metric balls generate a topology.
- Explain why equivalent finite-dimensional norms give the same filters of neighborhoods.

Practice problems.

1. Prove directly that the discrete metric on a set X induces the discrete topology. Describe the neighborhood filter $\mathcal{N}_X(x)$.
2. Let d and ρ be metrics on X such that $cd(x,y) \le \rho(x,y) \le Cd(x,y)$ for some $c, C > 0$. Prove that they induce the same topology and the same neighborhood filters.
3. On $\mathbb{R}^n$, prove the inequalities $d_\infty \le d_2 \le d_1 \le nd_\infty$ and use them to compare balls explicitly.
4. Give an example of two metrics with the same open sets but different Cauchy sequences. Explain which structure changed and which did not.

Ticket 2. Closure, limit points, and convergence

Core idea. Closure means that every neighborhood sees the set. Filters remove the need to choose a sequence unless the space has a countable local base.

Definition 2.1 (Closure and interior)**.** *For $A \subset X$, the closure of A is*

$$\overline{A} = \bigcap\{F \subset X : F \text{ is closed and } A \subset F\}.$$

The interior is

$$\operatorname{Int} A = \{x \in A : A \in \mathcal{N}_X(x)\}.$$

The boundary is $\partial A = \overline{A} \setminus \operatorname{Int} A$.

Proposition 2.1 (Neighborhood criterion for closure)**.** *For a topological space X, $A \subset X$, and $x \in X$,*

$$x \in \overline{A} \quad\Longleftrightarrow\quad \forall U \in \mathcal{N}_X(x),\ U \cap A \neq \varnothing.$$

Proof. If some neighborhood $U \in \mathcal{N}_X(x)$ is disjoint from A, choose an open O with $x \in O \subset U$. Then $X \setminus O$ is closed, contains A, and does not contain x. Hence $x \notin \overline{A}$.

Conversely, if $x \notin \overline{A}$, then $X \setminus \overline{A}$ is an open neighborhood of x disjoint from A. Thus not every neighborhood meets A. □

Definition 2.2 (Filter base attached to closure)**.** *If $x \in \overline{A}$, then*

$$\mathcal{B}_{A,x} = \{U \cap A : U \in \mathcal{N}_X(x)\}$$

is a filter base on A. The generated filter is the filter of points of A approaching x.

Why it is a filter base. Each $U \cap A$ is nonempty by the closure criterion. If $U, V \in \mathcal{N}_X(x)$, then $U \cap V \in \mathcal{N}_X(x)$, and

$$(U \cap V) \cap A \subset (U \cap A) \cap (V \cap A).$$

So the finite-intersection condition for a filter base holds. □

Definition 2.3 (Limit point)**.** *A point $x \in X$ is a limit point of $A \subset X$ if every neighborhood of x meets $A \setminus \{x\}$:*

$$\forall U \in \mathcal{N}_X(x), \quad U \cap (A \setminus \{x\}) \neq \varnothing.$$

Equivalently, $x \in \overline{A \setminus \{x\}}$.

Definition 2.4 (Sequence filter)**.** *A sequence (x_n) in X generates the filter*

$$\mathcal{F}_{(x_n)} = \{A \subset X : \exists N\ \forall n \geq N,\ x_n \in A\}.$$

Thus $A \in \mathcal{F}_{(x_n)}$ means that the sequence is eventually in A.

Definition 2.5 (Filter convergence)**.** *A filter $\mathcal{F}$ on a topological space X converges to x, written $\mathcal{F} \to x$, if*

$$\mathcal{N}_X(x) \subset \mathcal{F}.$$

For a sequence, this gives

$$x_n \to x \quad\Longleftrightarrow\quad \mathcal{F}_{(x_n)} \to x.$$

Proposition 2.2 (Sequential closure in first-countable spaces)**.** *If X is first-countable, then*

$$x \in \overline{A} \quad \Longleftrightarrow \quad \exists a_n \in A \text{ such that } a_n \to x.$$

In particular, this holds in every metric space.

Proof. If $a_n \in A$ and $a_n \to x$, then every neighborhood of x contains all sufficiently large a_n, hence meets A. So $x \in \overline{A}$.

Conversely, assume $x \in \overline{A}$. Since X is first-countable, choose a countable neighborhood base (U_n) at x. Replace it by the decreasing base

$$W_n = U_1 \cap \cdots \cap U_n.$$

Since $x \in \overline{A}$, each $W_n \cap A$ is nonempty. Choose $a_n \in W_n \cap A$. For any neighborhood O of x, some $W_N \subset O$. Since the W_n are decreasing, $a_n \in O$ for all $n \geq N$. Hence $a_n \to x$. □

Common mistake. Sequences do not characterize closure in arbitrary topological spaces. Filters do: $x \in \overline{A}$ exactly means that there exists a filter on A converging to x.

Oral-exam minimum.

- State closure via neighborhoods.
- Define sequence filters and filter convergence.
- Prove sequential closure under first-countability.
- Explain why filters are strictly more general than sequences.

Practice problems.

1. Prove that $x \in \overline{A}$ iff there exists a filter $\mathcal{F}$ on A such that $\mathcal{F} \to x$ in X.
2. In a metric space, prove that $x \in \overline{A}$ iff there is a sequence (a_n) in A with $a_n \to x$. Identify exactly where first countability is used.
3. Let $A \subset X$. Prove that $x \in \partial A$ iff every neighborhood of x meets both A and $X \setminus A$.
4. Construct a topological space where sequential closure is strictly smaller than topological closure. State why filters fix the problem.

Ticket 3. Cauchy filters, completeness, and metric completeness

Core idea. Cauchy behavior is not purely topological. It belongs to the uniform structure induced by the metric.

Definition 3.1 (Metric uniformity)**.** *For a metric space (X, d), define*

$$D_\varepsilon^X = \{(x, y) \in X \times X : d(x, y) < \varepsilon\}.$$

The metric uniformity $\mathcal{U}_X$ is the filter on $X \times X$ generated by the sets D_ε^X, $\varepsilon > 0$. Its elements are called entourages.

Definition 3.2 (Cauchy filter)**.** *A filter $\mathcal{F}$ on a metric space (X, d) is Cauchy if for every $\varepsilon > 0$ there exists $A \in \mathcal{F}$ such that*

$$\operatorname{diam}(A) < \varepsilon.$$

Equivalently,

$$\forall \varepsilon > 0 \ \exists A \in \mathcal{F} : \quad A \times A \subset D_\varepsilon^X.$$

Proposition 3.1. *If $\mathcal{F} \to x$ in a metric space, then $\mathcal{F}$ is Cauchy.*

Proof. Let $\varepsilon > 0$. Since $\mathcal{F} \to x$, the ball $B(x, \varepsilon/2)$ belongs to $\mathcal{F}$. Its diameter is at most ε. Hence $\mathcal{F}$ is Cauchy. $\square$

Definition 3.3 (Completeness through filters)**.** *A metric space X is complete if every Cauchy filter on X converges.*

Theorem 3.1 (Equivalence with sequential completeness)**.** *For metric spaces, the following are equivalent:*

1. *every Cauchy sequence converges;*

2. *every Cauchy filter converges.*

Proof. Assume every Cauchy filter converges. If (x_n) is a Cauchy sequence, its sequence filter $\mathcal{F}_{(x_n)}$ is Cauchy. Therefore $\mathcal{F}_{(x_n)} \to x$ for some x, which is exactly $x_n \to x$.

Conversely, assume every Cauchy sequence converges. Let $\mathcal{F}$ be a Cauchy filter. For each n, choose $A_n \in \mathcal{F}$ with $\operatorname{diam}(A_n) < 1/n$. Put

$$B_n = A_1 \cap \cdots \cap A_n \in \mathcal{F}.$$

Then $\operatorname{diam}(B_n) < 1/n$ and $B_{n+1} \subset B_n$. Choose $x_n \in B_n$. If $m, n \geq N$, then $x_m, x_n \in B_N$, so $d(x_m, x_n) < 1/N$. Hence (x_n) is Cauchy, and by assumption $x_n \to x$ for some $x \in X$.

We prove $\mathcal{F} \to x$. Let $\varepsilon > 0$. Choose N such that $1/N < \varepsilon/2$ and $d(x_N, x) < \varepsilon/2$. For every $y \in B_N$,

$$d(y, x) \leq d(y, x_N) + d(x_N, x) < \varepsilon.$$

Thus $B_N \subset B(x, \varepsilon)$. Since $B_N \in \mathcal{F}$, upward closure gives $B(x, \varepsilon) \in \mathcal{F}$. Therefore $\mathcal{F} \to x$. $\square$

Proposition 3.2 (Closed subspaces of complete spaces)**.** *Let X be complete and $F \subset X$ closed. Then F is complete with the inherited metric. Conversely, if $F \subset X$ is complete with the inherited metric, then F is closed in X.*

Proof. If $\mathcal{F}$ is a Cauchy filter on F, then its upward extension to X is a Cauchy filter on X, hence converges to some $x \in X$. Since all members of the original filter live in F, the point x belongs to $\overline{F} = F$. Hence F is complete.

Conversely, suppose F is complete and $x \in \overline{F}$. The family $B(x, 1/n) \cap F$ generates a Cauchy filter on F, so it converges in F to some $y \in F$. But it also converges to x in X. Metric limits are unique, so $x = y \in F$. Hence F is closed. $\square$

Oral-exam minimum.

- Define the metric uniformity $\mathcal{U}_X$.

- Define Cauchy filters and compare them with Cauchy sequences.

- Prove that, in metric spaces, sequence completeness and filter completeness coincide.

- Explain why Cauchy behavior is uniform, not merely topological.

Practice problems.

1. Let (x_n) be a sequence in a metric space. Prove that (x_n) is Cauchy iff its sequence filter is a Cauchy filter.

2. Prove that every convergent filter in a metric space is Cauchy.

3. Show that a complete subspace of a metric space is closed. Then prove the converse under the hypothesis that the ambient space is complete.

4. Prove that $\mathbb{R}^n$ is complete by using coordinate projections and the max metric.

Ticket 4. Compactness, ultrafilters, and closed-set principles

Core idea. Compactness says that no consistent system of closed constraints can escape the space. Ultrafilters express the same idea as convergence of maximal "eventually" structures.

Definition 4.1 (Compactness)**.** *A topological space X is compact if every open cover of X has a finite subcover.*

Definition 4.2 (Finite intersection property)**.** *A family $\{F_\alpha\}_{\alpha\in I}$ has the finite intersection property if every finite intersection is nonempty:*

$$F_{\alpha_1} \cap \cdots \cap F_{\alpha_n} \neq \varnothing.$$

Theorem 4.1 (Closed-set form of compactness)**.** *A topological space X is compact if and only if every family of closed subsets with the finite intersection property has nonempty total intersection.*

Proof. Let $F_\alpha = X \setminus U_\alpha$. The family $\{U_\alpha\}$ is an open cover with no finite subcover exactly when $\{F_\alpha\}$ is a family of closed sets with the finite intersection property and empty total intersection. Taking complements proves the equivalence. □

Definition 4.3 (Ultrafilter)**.** *A filter $\mathcal{U}$ on X is an ultrafilter if it is maximal among proper filters: there is no proper filter $\mathcal{G}$ with*

$$\mathcal{U} \subsetneq \mathcal{G}.$$

Equivalently, for every $A \subset X$, exactly one of the following holds:

$$A \in \mathcal{U}, \qquad X \setminus A \in \mathcal{U}.$$

Example 4.1 (Principal ultrafilter). For $x \in X$,

$$\mathcal{U}_x = \{A \subset X : x \in A\}$$

is an ultrafilter. It converges to x in every topology on X.

Theorem 4.2 (Ultrafilter compactness criterion)**.** *A topological space X is compact if and only if every ultrafilter on X converges to at least one point of X.*

Proof. Assume X is compact, and let $\mathcal{U}$ be an ultrafilter on X. If $\mathcal{U}$ does not converge to any point, then for each $x \in X$ choose a neighborhood $U_x \in \mathcal{N}_X(x)$ such that $U_x \notin \mathcal{U}$. Since $\mathcal{U}$ is an ultrafilter, $X \setminus U_x \in \mathcal{U}$. The sets U_x cover X. By compactness, finitely many cover X:

$$X = U_{x_1} \cup \cdots \cup U_{x_n}.$$

Then

$$\varnothing = (X \setminus U_{x_1}) \cap \cdots \cap (X \setminus U_{x_n}) \in \mathcal{U},$$

a contradiction.

Conversely, suppose every ultrafilter converges. If X is not compact, then there is a family of closed sets $\{F_\alpha\}$ with the finite intersection property and empty intersection. The finite intersections of the F_α's form a filter base; extend the generated filter to an ultrafilter $\mathcal{U}$. By assumption $\mathcal{U} \to x$ for some $x \in X$. Since each $F_\alpha \in \mathcal{U}$ and F_α is closed, convergence forces $x \in F_\alpha$ for every α. Thus $x \in \bigcap_\alpha F_\alpha$, contradiction. □

Proposition 4.1 (Finite products of compact spaces)**.** *If X and Y are compact, then $X \times Y$ is compact.*

Proof. Let $\mathcal{U}$ be an ultrafilter on $X \times Y$. The projection image filters $(\pi_X)_*\mathcal{U}$ and $(\pi_Y)_*\mathcal{U}$ extend to ultrafilters and therefore converge to some $x \in X$ and $y \in Y$. For a basic neighborhood $O \times V$ of (x, y), the sets $\pi_X^{-1}(O)$ and $\pi_Y^{-1}(V)$ belong to $\mathcal{U}$, so their intersection $O \times V$ belongs to $\mathcal{U}$. Hence $\mathcal{U} \to (x, y)$. By the ultrafilter criterion, $X \times Y$ is compact. □

Oral-exam minimum.

- Prove the closed-set compactness criterion.
- Define ultrafilter and principal ultrafilter.
- Prove compactness iff every ultrafilter converges.
- Use ultrafilters to prove finite products of compact spaces are compact.

Practice problems.

1. Prove that an ultrafilter $\mathcal{U}$ has the dichotomy property: for every $A \subset X$, exactly one of A and $X \setminus A$ belongs to $\mathcal{U}$.
2. Prove that the closed-set finite intersection property is equivalent to open-cover compactness.
3. Prove that a topological space is compact iff every ultrafilter on it converges.
4. Let $f : X \to Y$ be continuous and X compact. Prove $f(X)$ is compact using ultrafilters only.

Ticket 5. Total boundedness and compact metric spaces

Core idea. In metric spaces, compactness is equivalent to completeness plus total boundedness. Filters give a concise proof: every ultrafilter becomes Cauchy and then converges.

Definition 5.1 (Total boundedness)**.** *A metric space (X, d) is totally bounded if for every $\varepsilon > 0$ there exist finitely many points $x_1, \ldots, x_N \in X$ such that*

$$X \subset \bigcup_{j=1}^{N} B(x_j, \varepsilon).$$

For a subset $A \subset X$, this is applied to the inherited metric on A.

Proposition 5.1. *Every totally bounded metric space is bounded. The converse is false in general.*

Proof. If $X \subset \bigcup_{j=1}^{N} B(x_j, 1)$, then all of X lies in a sufficiently large ball around x_1, because the finite set $\{x_1, \ldots, x_N\}$ is bounded. The converse fails in infinite-dimensional normed spaces: in ℓ^∞, the unit vectors are bounded but pairwise distance 1, so no finite 1/3-net exists. □

Lemma 5.1 (Ultrafilters in totally bounded spaces are Cauchy)**.** *If X is totally bounded, then every ultrafilter on X is Cauchy.*

Proof. Let $\mathcal{U}$ be an ultrafilter and let $\varepsilon > 0$. Choose a finite cover

$$X \subset B(x_1, \varepsilon/2) \cup \cdots \cup B(x_N, \varepsilon/2).$$

Since $\mathcal{U}$ is an ultrafilter, one of these balls belongs to $\mathcal{U}$; otherwise all their complements would belong to $\mathcal{U}$, and their finite intersection would be empty. That ball has diameter at most ε. Hence $\mathcal{U}$ is Cauchy. □

Theorem 5.1 (Compact metric criterion). *For a metric space X,*

$$X \text{ compact} \iff X \text{ complete and totally bounded.}$$

Proof. Assume X is compact. To prove total boundedness, suppose not. Then for some $\varepsilon > 0$, no finite ε-net exists. Inductively choose x_{n+1} outside $B(x_1, \varepsilon) \cup \cdots \cup B(x_n, \varepsilon)$. Then $d(x_m, x_n) \geq \varepsilon$ for $m \neq n$. The sequence filter can be extended to an ultrafilter $\mathcal{U}$. Compactness gives $\mathcal{U} \to x$, hence $\mathcal{U}$ is Cauchy. But the separated construction prevents any set in the tail filter from having diameter $< \varepsilon$, contradiction. Thus X is totally bounded. Completeness follows because every Cauchy filter extends to an ultrafilter, and compactness makes that ultrafilter converge; the original Cauchy filter, being coarser than a convergent filter and Cauchy, converges to the same limit.

Conversely, assume X is complete and totally bounded. Let $\mathcal{U}$ be an ultrafilter on X. By total boundedness, $\mathcal{U}$ is Cauchy. By completeness, $\mathcal{U}$ converges. Hence every ultrafilter converges, so X is compact. □

Corollary 5.1 (Heine–Borel in $\mathbb{R}^n$). *A subset $A \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. A closed subset of the complete space $\mathbb{R}^n$ is complete. A bounded subset of $\mathbb{R}^n$ is totally bounded by subdivision into finitely many small cubes. Hence closed and bounded implies compact. Conversely, compact metric spaces are complete and totally bounded; in particular they are bounded and closed in any ambient metric space. □

Oral-exam minimum.

- Define total boundedness.
- Prove total boundedness implies boundedness and give an infinite-dimensional counterexample to the converse.
- Prove complete plus totally bounded implies compact using ultrafilters.
- Deduce Heine–Borel in $\mathbb{R}^n$.

Practice problems.

1. Prove that total boundedness implies boundedness in every metric space.
2. Give a bounded subset of an infinite-dimensional normed space that is not totally bounded, and prove it.
3. Prove that every bounded subset of $\mathbb{R}^n$ is totally bounded.
4. Prove that a compact metric space is totally bounded using an open cover by ε-balls.

Ticket 6. Sequential compactness as a metric shadow of compactness

Core idea. In metric spaces, compactness and sequential compactness coincide because first-countability lets filters produce sequences and total boundedness lets sequences produce Cauchy subsequences.

Definition 6.1 (Sequential compactness). *A metric space K is sequentially compact if every sequence in K has a convergent subsequence whose limit belongs to K.*

Theorem 6.1 (Compact metric implies sequentially compact). *Every compact metric space is sequentially compact.*

Proof. Let (x_n) be a sequence in compact K. Its sequence filter $\mathcal{F}_{(x_n)}$ extends to an ultrafilter $\mathcal{U}$. Since K is compact, $\mathcal{U} \to x$ for some $x \in K$.

We construct a subsequence converging to x. Since $\mathcal{U} \to x$, each ball $B(x, 1/k)$ belongs to $\mathcal{U}$. Each tail $\{x_n : n \geq N\}$ also belongs to $\mathcal{U}$. Therefore their intersection is nonempty. Choose recursively

$$n_k > n_{k-1}, \qquad x_{n_k} \in B(x, 1/k).$$

Then $x_{n_k} \to x$. Hence K is sequentially compact. □

Theorem 6.2 (Sequentially compact metric implies compact). *Every sequentially compact metric space is compact.*

Proof. First, K is totally bounded. If not, for some $\varepsilon > 0$ one can choose a sequence (x_n) with $d(x_m, x_n) \geq \varepsilon$ for $m \neq n$. Such a sequence has no Cauchy subsequence and hence no convergent subsequence, contradiction.

Second, K is complete. Let (x_n) be Cauchy. By sequential compactness it has a subsequence (x_{n_j}) converging to $x \in K$. Since the whole sequence is Cauchy, the whole sequence converges to the same x. Therefore K is complete.

By the compact metric criterion, complete plus totally bounded implies compact. □

Corollary 6.1. *For metric spaces,*

$$\boxed{\textit{compact} \Longleftrightarrow \textit{sequentially compact} \Longleftrightarrow \textit{complete and totally bounded.}}$$

Lemma 6.1 (Lebesgue number lemma). *Let K be compact metric, and let $\{U_\alpha\}$ be an open cover of K. Then there exists $\lambda > 0$ such that every subset of K with diameter less than λ is contained in some U_α.*

Proof. Suppose not. For each n, choose $A_n \subset K$ with $\operatorname{diam}(A_n) < 1/n$ such that A_n is contained in no member of the cover. Pick $x_n \in A_n$. By compactness, pass to a subsequence $x_{n_j} \to x \in K$. Choose α and $r > 0$ such that $B(x, r) \subset U_\alpha$. For large j, $d(x_{n_j}, x) < r/2$ and $1/n_j < r/2$. If $y \in A_{n_j}$, then

$$d(y, x) \leq d(y, x_{n_j}) + d(x_{n_j}, x) < r.$$

Thus $A_{n_j} \subset U_\alpha$, contradiction. □

Oral-exam minimum.

- Prove compact metric implies sequentially compact using ultrafilters.
- Prove sequentially compact metric implies complete and totally bounded.
- State the compact metric equivalence triangle.
- Prove the Lebesgue number lemma.

Practice problems.

1. Prove that compact metric spaces are sequentially compact using cluster points.
2. Prove that sequentially compact metric spaces are totally bounded.
3. Prove the Lebesgue number lemma for sequentially compact metric spaces.
4. Combine the previous results to prove the metric equivalence: compact iff sequentially compact iff complete and totally bounded.

Ticket 7. Limits and continuous mappings through filters

Core idea. A function is continuous if it sends the neighborhood filter of a point to a filter converging to the image point.

Definition 7.1 (Image filter). *Let $f : X \to Y$, and let $\mathcal{F}$ be a filter on X. The image filter is*

$$f_*\mathcal{F} = \{B \subset Y : f^{-1}(B) \in \mathcal{F}\}.$$

Definition 7.2 (Limit along a filter). *Let Y be topological. We say*

$$\lim_{\mathcal{F}} f = b$$

if

$$f_*\mathcal{F} \to b,$$

or equivalently,

$$\forall V \in \mathcal{N}_Y(b), \qquad f^{-1}(V) \in \mathcal{F}.$$

Definition 7.3 (Punctured limit along a subset). *Let $A \subset X$, $a \in \overline{A}$, and $f : A \to Y$. The punctured neighborhood filter of a along A is*

$$\mathcal{N}_A^\circ(a) = \{U \subset A : \exists W \in \mathcal{N}_X(a),\ (W \cap A) \setminus \{a\} \subset U\}.$$

Then

$$\lim_{x \to a,\ x \in A,\ x \neq a} f(x) = b$$

means

$$\lim_{\mathcal{N}_A^\circ(a)} f = b.$$

In metric spaces this is exactly

$$\forall \varepsilon > 0\ \exists \delta > 0 : \quad x \in A,\ 0 < d(x, a) < \delta \Rightarrow \rho(f(x), b) < \varepsilon.$$

Theorem 7.1 (Continuity at a point). *For $f : X \to Y$,*

$$f \text{ is continuous at } x \quad \Longleftrightarrow \quad f_*\mathcal{N}_X(x) \to f(x).$$

Equivalently,

$$\mathcal{N}_Y(f(x)) \subset f_*\mathcal{N}_X(x).$$

Proof. The condition $f_*\mathcal{N}_X(x) \to f(x)$ means

$$\forall V \in \mathcal{N}_Y(f(x)), \quad V \in f_*\mathcal{N}_X(x).$$

By definition of image filter,

$$V \in f_*\mathcal{N}_X(x) \quad \Longleftrightarrow \quad f^{-1}(V) \in \mathcal{N}_X(x).$$

Thus the condition becomes

$$\forall V \in \mathcal{N}_Y(f(x)), \quad f^{-1}(V) \in \mathcal{N}_X(x),$$

which is exactly the topological definition of continuity at x. □

Theorem 7.2 (Continuity preserves all convergent filters)**.** *A map $f : X \to Y$ is continuous if and only if for every filter $\mathcal{F}$ on X,*

$$\mathcal{F} \to x \quad \Longrightarrow \quad f_*\mathcal{F} \to f(x).$$

Proof. If f is continuous and $\mathcal{F} \to x$, let $V \in \mathcal{N}_Y(f(x))$. Then $f^{-1}(V) \in \mathcal{N}_X(x)$. Since $\mathcal{N}_X(x) \subset \mathcal{F}$, we get $f^{-1}(V) \in \mathcal{F}$. Hence $V \in f_*\mathcal{F}$, so $f_*\mathcal{F} \to f(x)$.

Conversely, apply the stated property to $\mathcal{F} = \mathcal{N}_X(x)$. Then $f_*\mathcal{N}_X(x) \to f(x)$, so f is continuous at x. Since x is arbitrary, f is continuous. □

Theorem 7.3 (Heine criterion in first-countable spaces)**.** *If X is first-countable, then $f : X \to Y$ is continuous at x if and only if*

$$x_n \to x \quad \Longrightarrow \quad f(x_n) \to f(x)$$

for every sequence (x_n) in X.

Proof. Continuity implies the sequential statement because sequence filters are filters.

Conversely, assume the sequential statement holds but f is not continuous at x. Then for some $V \in \mathcal{N}_Y(f(x))$, the preimage $f^{-1}(V)$ is not a neighborhood of x. Let (W_n) be a decreasing countable local base at x. For each n, choose

$$x_n \in W_n \setminus f^{-1}(V).$$

Then $x_n \to x$, but $f(x_n) \notin V$ for all n, so $f(x_n) \not\to f(x)$. Contradiction. □

Oral-exam minimum.

- Define image filters.
- Write continuity at x as $f_*\mathcal{N}_X(x) \to f(x)$.
- Prove the preservation-of-filters criterion.
- Explain why the sequence criterion needs first-countability.

Practice problems.

1. Let $f : X \to Y$. Prove that f is continuous at x iff $f_*\mathcal{N}_X(x) \to f(x)$.

2. Prove that f is continuous iff for every filter $\mathcal{F} \to x$, one has $f_*\mathcal{F} \to f(x)$.

3. In metric spaces, expand the filter definition of $\lim_{x\to a} f(x) = b$ into the usual punctured ε-δ definition.

4. Prove that in first-countable spaces sequential continuity implies continuity. Then give a reason why this can fail in general topological spaces.

Ticket 8. Connectedness and continuous images

Core idea. Connectedness is purely topological: it is exactly the impossibility of separating the space into two nonempty open pieces.

Definition 8.1 (Separation and connectedness)**.** *A topological space X is disconnected if there exist nonempty disjoint open sets $U, V \subset X$ such that*

$$X = U \cup V.$$

If no such pair exists, X is connected. A subset $A \subset X$ is connected if it is connected with the subspace topology.

Theorem 8.1 (Continuous images preserve connectedness)**.** *If X is connected and $f : X \to Y$ is continuous, then $f(X)$ is connected.*

Proof. Suppose $f(X) = A \cup B$ is a separation in the subspace topology of $f(X)$. Then

$$f^{-1}(A), \qquad f^{-1}(B)$$

are disjoint nonempty open subsets of X whose union is X. This contradicts connectedness of X. □

Theorem 8.2 (Intervals are connected)**.** *Every interval $I \subset \mathbb{R}$ is connected.*

Proof. Assume $I = A \cup B$ is a separation. Choose $a \in A$, $b \in B$, with $a < b$. Let

$$c = \sup(A \cap [a, b]).$$

Since I is an interval, $c \in I$. If $c \in A$, relative openness of A gives points of A to the right of c, contradicting the definition of supremum. If $c \in B$, relative openness of B gives a left neighborhood of c contained in B, contradicting that c is the supremum of points of $A \cap [a, b]$. Hence no separation exists. □

Theorem 8.3 (Intermediate value theorem)**.** *If K is connected and $f : K \to \mathbb{R}$ is continuous, then $f(K)$ is an interval. If additionally K is compact, then*

$$f(K) = [m, M],$$

where $m = \min_K f$ and $M = \max_K f$.

Proof. The image $f(K)$ is connected, hence an interval in $\mathbb{R}$. If K is compact, then $f(K)$ is compact, hence closed and bounded. Therefore it contains its infimum and supremum. □

Oral-exam minimum.

- Define connectedness through separations.
- Prove continuous images preserve connectedness.
- Prove intervals in $\mathbb{R}$ are connected.
- Derive the intermediate value theorem.

Practice problems.

1. Prove that X is connected iff every continuous map $X \to \{0, 1\}$, where $\{0, 1\}$ has the discrete topology, is constant.
2. Prove that the continuous image of a connected space is connected.
3. Prove that every interval in $\mathbb{R}$ is connected using the supremum argument.
4. Let K be compact and connected, and $f : K \to \mathbb{R}$ continuous. Prove that $f(K) = [m, M]$ for some $m, M \in \mathbb{R}$.

Ticket 9. Uniform continuity and Cantor theorem

Core idea. Uniform continuity is not pointwise. It says that closeness to the diagonal in $X \times X$ is sent to closeness to the diagonal in $Y \times Y$.

Definition 9.1 (Uniform continuity). *For metric spaces (X, d) and (Y, ρ), a map $f : X \to Y$ is uniformly continuous if*

$$\forall \varepsilon > 0 \ \exists \delta > 0 : \quad d(x, y) < \delta \Rightarrow \rho(f(x), f(y)) < \varepsilon.$$

Theorem 9.1 (Filter/uniformity form of uniform continuity). *Let $\mathcal{U}_X$ and $\mathcal{U}_Y$ be the metric uniformities. Then $f : X \to Y$ is uniformly continuous if and only if*

$$\forall V \in \mathcal{U}_Y, \qquad (f \times f)^{-1}(V) \in \mathcal{U}_X.$$

Equivalently,

$$\mathcal{U}_Y \subset (f \times f)_* \mathcal{U}_X.$$

Proof. It is enough to test the basic entourages

$$D^Y_\varepsilon = \{(u, v) : \rho(u, v) < \varepsilon\}.$$

The condition

$$(f \times f)^{-1}(D^Y_\varepsilon) \in \mathcal{U}_X$$

means that for some $\delta > 0$,

$$D^X_\delta \subset (f \times f)^{-1}(D^Y_\varepsilon).$$

Expanding this inclusion gives exactly

$$d(x, y) < \delta \Rightarrow \rho(f(x), f(y)) < \varepsilon.$$

□

Theorem 9.2 (Cantor theorem / Heine–Cantor). *Let K be a compact metric space and let Y be a metric space. If $f : K \to Y$ is continuous, then f is uniformly continuous.*

Filter proof. Assume f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$, there are $x, y \in K$ satisfying

$$d(x, y) < \delta, \qquad \rho(f(x), f(y)) \geq \varepsilon_0.$$

Let

$$D^K_\delta = \{(x, y) \in K \times K : d(x, y) < \delta\}$$

and define the bad set

$$C = \{(x, y) \in K \times K : \rho(f(x), f(y)) \geq \varepsilon_0\}.$$

The failure of uniform continuity says

$$D^K_\delta \cap C \neq \varnothing \quad \text{for every } \delta > 0.$$

Thus the family $\{D^K_\delta \cap C : \delta > 0\}$ is a filter base on $K \times K$. Let $\mathcal{F}$ be the generated filter. Then

$$C \in \mathcal{F}, \qquad D^K_\delta \in \mathcal{F} \quad (\delta > 0).$$

Extend $\mathcal{F}$ to an ultrafilter $\mathcal{G}$ on $K \times K$. Since K is compact, $K \times K$ is compact, so $\mathcal{G} \to (a, b)$ for some $(a, b) \in K \times K$.

Because $D^K_\delta \in \mathcal{G}$ for every $\delta > 0$, the limit must lie on the diagonal. Indeed, if $a \neq b$, choose $\delta < d(a,b)/3$. A sufficiently small product neighborhood of (a,b) is disjoint from D^K_δ, contradicting the fact that both sets belong to $\mathcal{G}$. Hence $a = b$, so $\mathcal{G} \to (a,a)$.

Continuity of f implies continuity of $f \times f$. Therefore

$$(f \times f)_*\mathcal{G} \to (f(a), f(a)).$$

The set

$$D^Y_{\varepsilon_0} = \{(u,v) : \rho(u,v) < \varepsilon_0\}$$

is a neighborhood of $(f(a), f(a))$, hence

$$(f \times f)^{-1}(D^Y_{\varepsilon_0}) \in \mathcal{G}.$$

But by definition of C,

$$C = (K \times K) \setminus (f \times f)^{-1}(D^Y_{\varepsilon_0}).$$

Since also $C \in \mathcal{G}$, the filter $\mathcal{G}$ contains two disjoint sets. Hence it contains $\varnothing$, impossible. Therefore f is uniformly continuous. $\square$

Oral-exam minimum.

- State uniform continuity through $(f \times f)^{-1}$.
- Explain why uniform continuity requires a uniformity, not just a topology.
- Prove Cantor theorem using ultrafilters on $K \times K$.
- Identify exactly where compactness is used.

Practice problems.

1. Define the metric uniformity $\mathcal{U}_X$ and prove that it is a filter on $X \times X$.
2. Prove that $f : X \to Y$ is uniformly continuous iff $(f \times f)^{-1}(V) \in \mathcal{U}_X$ for every $V \in \mathcal{U}_Y$.
3. Prove Cantor's theorem using ultrafilters: if K is compact metric and $f : K \to Y$ is continuous, then f is uniformly continuous.
4. Give an example of a continuous function on a noncompact metric space that is not uniformly continuous. Identify where the compactness proof fails.

Ticket 10. Pointwise and uniform convergence of functions

Core idea. Pointwise convergence is convergence after every evaluation map. Uniform convergence is convergence in a function-space uniformity.

Definition 10.1 (Pointwise convergence)**.** *Let E be a set and (Y, ρ) a metric space. A sequence $f_n : E \to Y$ converges pointwise to $f : E \to Y$ if*

$$\forall x \in E, \qquad f_n(x) \to f(x).$$

Equivalently, for every evaluation map

$$\mathrm{ev}_x : Y^E \to Y, \qquad \mathrm{ev}_x(g) = g(x),$$

we have $\mathrm{ev}_x(f_n) \to \mathrm{ev}_x(f)$.

Definition 10.2 (Uniform convergence)**.** *If $Y = \mathbb{K}$ or more generally if Y is metric, $f_n : E \to Y$ converges uniformly to f if*

$$\forall \varepsilon > 0\ \exists N\ \forall n \geq N\ \forall x \in E : \rho(f_n(x), f(x)) < \varepsilon.$$

For scalar bounded functions this is convergence in the sup metric

$$d_\infty(f, g) = \sup_{x \in E} |f(x) - g(x)|.$$

Theorem 10.1 (Uniform Cauchy criterion)**.** *Let Y be complete. A sequence $f_n : E \to Y$ converges uniformly to some $f : E \to Y$ if and only if*

$$\forall \varepsilon > 0\ \exists N\ \forall m, n \geq N\ \forall x \in E : \rho(f_m(x), f_n(x)) < \varepsilon.$$

Proof. Uniform convergence implies the criterion by the triangle inequality. Conversely, the criterion makes $(f_n(x))$ Cauchy in Y for every fixed x. Since Y is complete, define

$$f(x) = \lim_n f_n(x).$$

Fix $\varepsilon > 0$. Choose N such that $\rho(f_m(x), f_n(x)) < \varepsilon$ for all $m, n \geq N$ and all x. Letting $m \to \infty$ gives

$$\rho(f(x), f_n(x)) \leq \varepsilon$$

for every x and all $n \geq N$. Hence $f_n \to f$ uniformly. □

Definition 10.3 (Uniform convergence of series)**.** *A series $\sum u_n(x)$ converges uniformly if its partial sums*

$$S_N(x) = \sum_{n=1}^{N} u_n(x)$$

converge uniformly.

Theorem 10.2 (Weierstrass M-test)**.** *If $|u_n(x)| \leq M_n$ for all $x \in E$ and $\sum M_n < \infty$, then $\sum u_n$ converges absolutely and uniformly on E.*

Proof. For $m > n$,

$$\sup_{x \in E} \left| \sum_{k=n+1}^{m} u_k(x) \right| \leq \sum_{k=n+1}^{m} M_k.$$

The right side tends to 0 as $m, n \to \infty$. The uniform Cauchy criterion applies. □

Oral-exam minimum.

- Distinguish pointwise from uniform convergence.
- Express pointwise convergence using evaluation maps.
- Prove the uniform Cauchy criterion.
- Prove the Weierstrass M-test.

Practice problems.

1. For $f_n : E \to \mathbb{K}$, prove that uniform convergence is exactly convergence in the sup metric whenever the suprema are finite.
2. Prove the uniform Cauchy criterion for sequences of functions $f_n : E \to \mathbb{K}$.
3. Show that $f_n(x) = x^n$ on $[0, 1]$ converges pointwise but not uniformly.
4. Prove the Weierstrass M-test using the uniform Cauchy criterion.

Ticket 11. Continuity of uniform limits

Core idea. Uniform convergence lets one control the limit function globally, while continuity controls one finite approximating function locally.

Theorem 11.1 (Uniform limit theorem)**.** *Let X be a topological space and (Y, ρ) a metric space. If $f_n : X \to Y$ are continuous and $f_n \to f$ uniformly, then f is continuous.*

Proof. Fix $x_0 \in X$ and $\varepsilon > 0$. Choose N such that

$$\rho(f_N(x), f(x)) < \varepsilon/3$$

for all $x \in X$. By continuity of f_N at x_0, the set

$$U = f_N^{-1}(B(f_N(x_0), \varepsilon/3))$$

is a neighborhood of x_0. If $x \in U$, then

$$\rho(f(x), f(x_0)) \le \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(x_0)) + \rho(f_N(x_0), f(x_0)) < \varepsilon.$$

Thus $f^{-1}(B(f(x_0), \varepsilon)) \in \mathcal{N}_X(x_0)$, so f is continuous at x_0. □

Filter formulation. Let $\mathcal{F} \to x_0$. We must prove $f_*\mathcal{F} \to f(x_0)$. Given $\varepsilon > 0$, choose N as above. Since f_N is continuous,

$$(f_N)_*\mathcal{F} \to f_N(x_0).$$

Hence

$$f_N^{-1}(B(f_N(x_0), \varepsilon/3)) \in \mathcal{F}.$$

On this set, the same triangle estimate gives $f(x) \in B(f(x_0), \varepsilon)$. Therefore $f^{-1}(B(f(x_0), \varepsilon)) \in \mathcal{F}$, proving $f_*\mathcal{F} \to f(x_0)$. □

Corollary 11.1. *If $u_n : X \to \mathbb{K}$ are continuous and $\sum u_n$ converges uniformly, then its sum is continuous.*

Oral-exam minimum.

- Prove the uniform limit theorem with the $\varepsilon/3$ estimate.
- Rewrite the proof using convergent filters.
- Explain why pointwise convergence is not enough.

Practice problems.

1. Prove that if $f_n : X \to Y$ are continuous, Y is metric, and $f_n \to f$ uniformly, then f is continuous.
2. Rewrite the proof of uniform-limit continuity using neighborhood filters instead of $\varepsilon/3$ notation.
3. Give an example of pointwise convergence of continuous functions to a discontinuous function.
4. Prove that if $\sum u_n$ converges uniformly and every u_n is continuous, then the sum is continuous.

Ticket 12. Integrals and derivatives as continuous operations

Core idea. A limit passes through an integral because the integral is a continuous linear functional on the normed space $C([a,b])$.

Definition 12.1 (Normed vector space)**.** *A normed vector space is a vector space V over $\mathbb{K}$ with a norm $\|\cdot\|$. A linear map $L : V \to \mathbb{K}$ is bounded if there exists $C \geq 0$ such that*

$$|L(v)| \leq C\|v\| \quad (v \in V).$$

Proposition 12.1. *Every bounded linear functional is continuous.*

Proof. If $v_i \to v$, then

$$|L(v_i) - L(v)| = |L(v_i - v)| \leq C\|v_i - v\| \to 0.$$

In filter language, if $\mathcal{F} \to v$, then for every ball around $L(v)$, the inverse image contains a ball around v, hence belongs to $\mathcal{F}$. Thus $L_*\mathcal{F} \to L(v)$. □

Proposition 12.2 (Integral functional)**.** *On $C([a,b])$ with the uniform norm*

$$\|f\|_\infty = \sup_{x\in[a,b]} |f(x)|,$$

define

$$I(f) = \int_a^b f(x)\,dx.$$

Then I is a bounded linear functional and

$$|I(f)| \leq (b-a)\|f\|_\infty.$$

Proof. Linearity is the linearity of the Riemann integral. The estimate follows from

$$\left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx \leq (b-a)\|f\|_\infty.$$

□

Theorem 12.1 (Limit under the integral sign)**.** *If $f_n \in C([a,b])$ and $f_n \to f$ uniformly, then*

$$\int_a^b f_n(x)\,dx \to \int_a^b f(x)\,dx.$$

Proof. Uniform convergence is exactly $\|f_n - f\|_\infty \to 0$. Hence

$$\left|\int_a^b f_n - \int_a^b f\right| = |I(f_n - f)| \leq (b-a)\|f_n - f\|_\infty \to 0.$$

Equivalently, this is simply continuity of the map $I : C([a,b]) \to \mathbb{K}$. □

Corollary 12.1 (Termwise integration)**.** *If $u_n \in C([a,b])$ and $\sum u_n$ converges uniformly to S, then*

$$\int_a^b S(x)\,dx = \sum_{n=1}^{\infty} \int_a^b u_n(x)\,dx.$$

Oral-exam minimum.

- Define bounded linear functional.
- Prove bounded linear functionals are continuous.
- Prove the integral estimate $|I(f)| \leq (b-a)\|f\|_\infty$.
- Derive passage of limits through integrals.

12.1 Differentiating uniform limits

Core idea. Uniform convergence of derivatives plus convergence at one point reconstructs the functions by the fundamental theorem of calculus.

Theorem 12.2 (Differentiation theorem for uniform limits)**.** *Let $f_n \in C^1([a,b])$. Suppose $f_n(x_0)$ converges for some $x_0 \in [a,b]$, and $f_n' \to g$ uniformly on $[a,b]$. Then f_n converges uniformly to a function $f \in C^1([a,b])$, and*

$$f' = g.$$

Proof. By the fundamental theorem of calculus,

$$f_n(x) = f_n(x_0) + \int_{x_0}^{x} f_n'(t)\,dt.$$

Let $c = \lim_n f_n(x_0)$. Since $f_n' \to g$ uniformly,

$$\sup_{x\in[a,b]} \left| \int_{x_0}^{x} (f_n'(t) - g(t))\,dt \right| \le (b-a)\|f_n' - g\|_\infty \to 0.$$

Thus $f_n \to f$ uniformly, where

$$f(x) = c + \int_{x_0}^{x} g(t)\,dt.$$

Since g is continuous as a uniform limit of continuous functions, the fundamental theorem gives $f' = g$. $\square$

Corollary 12.2 (Termwise differentiation of series)**.** *Let $u_n \in C^1([a,b])$. If $\sum u_n(x_0)$ converges at one point and $\sum u_n'$ converges uniformly on $[a,b]$, then $\sum u_n$ converges uniformly to a differentiable function S, and*

$$S'(x) = \sum_{n=1}^{\infty} u_n'(x).$$

Oral-exam minimum.

- State the hypotheses: convergence at one point plus uniform convergence of derivatives.
- Prove uniform convergence of f_n using the integral formula.
- Explain why pointwise convergence of derivatives is not enough.

Practice problems.

1. Prove that $I : C([a,b]) \to \mathbb{K}$, $I(f) = \int_a^b f(x)\,dx$, is a bounded linear functional and compute a valid operator norm bound.
2. Use the continuity of I to prove that uniform convergence permits passage of limits through integrals.
3. Prove termwise integration of a uniformly convergent series of continuous functions.
4. Prove the theorem on differentiating a uniformly convergent sequence under the hypotheses: $f_n(x_0)$ converges and $f_n' \to g$ uniformly.

Ticket 13. Parameter integrals as maps into function spaces

Core idea. A parameter integral is a composition: the parameter selects a function, and a continuous linear functional integrates it.

Definition 13.1. *Let T be a topological space. A map*

$$G : T \to C([a, b])$$

is continuous at t_0 if

$$G_*\mathcal{N}_T(t_0) \to G(t_0)$$

in the uniform norm topology. If $G(t)(x) = f(x, t)$, this means uniform convergence in x as $t \to t_0$:

$$\sup_{x\in[a,b]} |f(x, t) - f(x, t_0)| \to 0.$$

Theorem 13.1 (Continuity of parameter integrals)**.** *Let $G : T \to C([a, b])$ be continuous at t_0. Define*

$$F(t) = \int_a^b G(t)(x)\, dx.$$

Then F is continuous at t_0.

Proof. Let $I : C([a, b]) \to \mathbb{K}$ be the integral functional. Then

$$F = I \circ G.$$

Since G is continuous at t_0 and I is continuous, the composition is continuous. Quantitatively,

$$|F(t) - F(t_0)| = |I(G(t) - G(t_0))| \le (b - a)\|G(t) - G(t_0)\|_\infty \to 0.$$

□

Corollary 13.1 (Continuous kernel on a compact rectangle)**.** *If $f : [a, b] \times T_0 \to \mathbb{K}$ is continuous and T_0 is compact metric, then*

$$F(t) = \int_a^b f(x, t)\, dx$$

is continuous on T_0.

Proof. The set $[a, b] \times T_0$ is compact. By Cantor theorem, f is uniformly continuous on it. Hence $t \to t_0$ implies

$$\sup_{x\in[a,b]} |f(x, t) - f(x, t_0)| \to 0.$$

Thus the map $G(t) = f(\cdot, t)$ is continuous into $C([a, b])$, and the previous theorem applies. □

Definition 13.2 (Differentiability into a function space)**.** *Let $J \subset \mathbb{R}$. A map $G : J \to C([a, b])$ is differentiable at $t \in J$ with derivative $H \in C([a, b])$ if*

$$\left\| \frac{G(t + h) - G(t)}{h} - H \right\|_\infty \to 0 \quad (h \to 0).$$

Theorem 13.2 (Differentiation under the integral sign)**.** *If $G : J \to C([a,b])$ is differentiable at t, then*

$$F(t) = \int_a^b G(t)(x)\,dx$$

is differentiable at t, and

$$F'(t) = \int_a^b G'(t)(x)\,dx.$$

Proof. Using linearity of I,

$$\frac{F(t+h) - F(t)}{h} - I(G'(t)) = I\left(\frac{G(t+h) - G(t)}{h} - G'(t)\right).$$

Taking absolute values and using boundedness of I, the absolute value is at most

$$(b-a)\left\|\frac{G(t+h) - G(t)}{h} - G'(t)\right\|_\infty \to 0.$$

□

Oral-exam minimum.

- Interpret $f(x,t)$ as a curve $G(t) = f(\cdot,t)$ in $C([a,b])$.
- Prove parameter continuity by composition with the integral functional.
- Prove differentiation under the integral sign by differentiability in norm.

Practice problems.

1. Let $G : T \to C([a,b])$ be continuous at t_0. Prove that $F(t) = \int_a^b G(t)(x)\,dx$ is continuous at t_0.
2. If $f : [a,b] \times T_0 \to \mathbb{K}$ is continuous and T_0 compact metric, prove that $t \mapsto f(\cdot,t)$ is continuous into $C([a,b])$.
3. Prove differentiation under the integral sign by treating $G : J \to C([a,b])$ as a differentiable curve in a normed space.
4. Work out an example where pointwise continuity in t is not enough to get continuity of $\int f(x,t)\,dx$ without some uniform control.

Ticket 14. Power series and local uniform convergence

Core idea. A power series has a radius inside which convergence is locally uniform. Local uniformity is what permits continuity, integration, and differentiation term by term.

Definition 14.1 (Power series)**.** *A complex power series centered at z_0 is*

$$\sum_{n=0}^{\infty} a_n(z-z_0)^n.$$

Its radius of convergence is $R \in [0,\infty]$ such that the series converges absolutely for $|z-z_0| < R$ and diverges for $|z-z_0| > R$.

Theorem 14.1 (Cauchy–Hadamard)**.** *For the power series* $\sum a_n(z-z_0)^n$*,*

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n},$$

with the conventions $1/0 = \infty$ *and* $1/\infty = 0$*.*

Proof. Let $L = \limsup |a_n|^{1/n}$. If $r < 1/L$, choose $q < 1$ with $rL < q$. For all large n, $|a_n|^{1/n} r < q$, hence $|a_n|r^n < q^n$. Therefore the series converges absolutely for $|z - z_0| \le r$.

If $r > 1/L$, then infinitely many n satisfy $|a_n|^{1/n} r > 1$, so $|a_n|r^n > 1$ infinitely often. The terms do not tend to zero, so the series diverges for $|z - z_0| = r$ along those radii. □

Theorem 14.2 (Local uniform convergence inside the disk)**.** *For every* $0 < r < R$*, the series*

$$\sum a_n(z-z_0)^n$$

converges absolutely and uniformly on the closed disk $|z - z_0| \le r$*.*

Proof. Choose s with $r < s < R$. Since $s < R$, the numerical series $\sum |a_n|s^n$ converges. For $|z - z_0| \le r$,

$$|a_n(z-z_0)^n| \le |a_n|r^n \le |a_n|s^n.$$

The Weierstrass M-test gives absolute uniform convergence. □

Remark 14.1 (Filter language)*.* For every compact disk $K_r = \{|z - z_0| \le r\} \subset \{|z - z_0| < R\}$, the partial sums form a Cauchy sequence in the Banach space $C(K_r)$ with the sup norm. Thus the power series is best understood as a locally uniform limit of polynomials.

Oral-exam minimum.

- State and prove Cauchy–Hadamard.
- Explain why convergence is locally uniform strictly inside the disk.
- Explain why no universal statement holds on the boundary $|z - z_0| = R$.

Practice problems.

1. Prove the Cauchy-Hadamard formula for the radius of convergence.

2. Prove that a power series converges locally uniformly inside its disk of convergence.

3. Give examples showing that no universal statement is possible on the boundary $|z - z_0| = R$.

4. Use the M-test to prove uniform convergence of $\sum a_n(z-z_0)^n$ on every closed disk $|z - z_0| \le r < R$.

Ticket 15. Termwise integration and differentiation of power series

Core idea. Inside every smaller closed disk, a power series is a uniform limit of polynomials with uniformly convergent derivative series.

Theorem 15.1 (Derivative radius is invariant)**.** *The derivative series*

$$\sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}$$

has the same radius of convergence as

$$\sum_{n=0}^{\infty} a_n (z-z_0)^n.$$

Proof. By Cauchy–Hadamard, the reciprocal radius of the derivative series is

$$\limsup_{n\to\infty} |na_n|^{1/n} = \Big(\lim_{n\to\infty} n^{1/n}\Big) \limsup_{n\to\infty} |a_n|^{1/n} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Thus the radii agree. □

Theorem 15.2 (Termwise differentiation)**.** *For $|z - z_0| < R$, the sum*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

is differentiable, and

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}.$$

Proof. Fix a closed disk $|z - z_0| \le r < R$. The original series and the derivative series converge uniformly on this disk by local uniform convergence and the invariant-radius theorem. Apply the termwise differentiation theorem for uniform limits on compact intervals in the real case, or the corresponding complex proof using difference quotients uniformly on smaller disks. Hence differentiation is termwise inside the convergence disk. □

Theorem 15.3 (Termwise integration)**.** *For real x with $|x - x_0| < R$,*

$$\int_{x_0}^{x} \sum_{n=0}^{\infty} a_n (t - x_0)^n \, dt = \sum_{n=0}^{\infty} a_n \frac{(x - x_0)^{n+1}}{n+1}.$$

The integrated series has the same radius R.

Proof. On every closed interval $[x_0 - r, x_0 + r] \subset (x_0 - R, x_0 + R)$, the power series converges uniformly. Therefore the limit may be passed through the integral. The radius of the integrated series is unchanged because

$$(n+1)^{1/n} \to 1.$$

□

Oral-exam minimum.

- Prove that the derivative series has the same radius.
- Prove termwise differentiation using uniform convergence of derivative series.
- Prove termwise integration using the integral functional.

Practice problems.

1. Prove that the derivative series $\sum_{n\ge 1} n a_n (z - z_0)^{n-1}$ has the same radius of convergence as the original power series.
2. Prove termwise differentiation of a real power series on compact subintervals inside its interval of convergence.
3. Prove termwise integration of a power series and show that the integrated series has the same radius.
4. Apply termwise integration to derive the series for $\log(1 + x)$ from the geometric series for $1/(1 + x)$.

Ticket 16. Analytic functions, Taylor coefficients, and elementary expansions

Core idea. Analyticity means local equality with a power series. The coefficients are forced by derivatives, so power-series representations are unique.

Definition 16.1 (Analyticity)**.** *A function f is analytic at x_0 if there exist $R > 0$ and coefficients (a_n) such that*

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \quad (|x - x_0| < R).$$

Theorem 16.1 (Uniqueness of power-series representation)**.** *If*

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = \sum_{n=0}^{\infty} b_n (x - x_0)^n$$

for all x in a neighborhood of x_0, then $a_n = b_n$ for every n.

Proof. Subtract the two series. It is enough to prove: if

$$\sum_{n=0}^{\infty} c_n (x - x_0)^n = 0$$

near x_0, then all $c_n = 0$. Substituting $x = x_0$ gives $c_0 = 0$. For $x \neq x_0$, divide by $x - x_0$:

$$\sum_{n=1}^{\infty} c_n (x - x_0)^{n-1} = 0.$$

Letting $x \to x_0$, local uniform convergence justifies passing to the limit and gives $c_1 = 0$. Repeating inductively gives all coefficients zero. □

Corollary 16.1 (Taylor coefficients)**.** *If*

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

near x_0, then

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

Proof. By termwise differentiation,

$$f^{(n)}(x) = \sum_{k=n}^{\infty} k(k-1)\cdots(k-n+1) a_k (x - x_0)^{k-n}.$$

Substitute $x = x_0$. Only the $k = n$ term remains:

$$f^{(n)}(x_0) = n! a_n.$$

□

Theorem 16.2 (Elementary series)**.** *For their usual radii of convergence,*

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!},$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \qquad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1),$$

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \quad (-1 < x \le 1).$$

Proof idea. The geometric series follows from finite geometric sums and passage to the limit. The logarithmic series follows by integrating the geometric series for $1/(1+x)$ term by term on compact subintervals of $(-1,1)$, with endpoint $x = 1$ handled by the alternating series theorem or Abel-type limiting. The exponential, sine, and cosine series follow from solving the differential equations $f' = f$, $s' = c$, $c' = -s$ with the corresponding initial values and using uniqueness of power-series solutions. □

Theorem 16.3 (Euler formula)**.** *For every real x,*

$$e^{ix} = \cos x + i \sin x.$$

Proof. Using the absolutely convergent power series for e^z,

$$e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = \cos x + i \sin x.$$

□

Oral-exam minimum.

- Define analyticity.
- Prove uniqueness of power-series coefficients.
- Derive Taylor coefficients from termwise differentiation.
- Prove Euler's formula using absolute convergence.

Practice problems.

1. Prove uniqueness of power-series coefficients by induction using division by $x - x_0$.
2. Prove that if $f(x) = \sum a_n (x - x_0)^n$ near x_0, then $a_n = f^{(n)}(x_0)/n!$.
3. Derive the Taylor series for e^x, $\sin x$, and $\cos x$ using differential equations and initial values.
4. Prove Euler's formula $e^{ix} = \cos x + i \sin x$ from the power-series definitions.

Final checklist

Before the exam, make sure you can do the following without notes:

1. Define metric topology and neighborhood filters.
2. Prove closure through neighborhoods and explain the filter base $U \cap A$.

3. Define image filters and write continuity as $f_*\mathcal{N}_X(x) \to f(x)$.
4. Prove continuity preserves all convergent filters.
5. Explain why sequences characterize continuity only under first-countability.
6. Define metric uniformity and Cauchy filters.
7. Prove Cauchy-filter completeness equals sequential completeness in metric spaces.
8. Define ultrafilters and prove the ultrafilter compactness criterion.
9. Prove compact metric $\iff$ complete plus totally bounded.
10. Prove compact metric $\iff$ sequentially compact.
11. Write uniform continuity as $\mathcal{U}_Y \subset (f \times f)_*\mathcal{U}_X$.
12. Prove Heine–Cantor using an ultrafilter on $K \times K$.
13. Prove the uniform Cauchy criterion and Weierstrass M-test.
14. Prove uniform limits of continuous functions are continuous using filters.
15. Treat integration as a bounded linear functional on $C([a, b])$.
16. Treat parameter integrals as compositions $T \to C([a, b]) \to \mathbb{K}$.
17. Prove Cauchy–Hadamard and local uniform convergence of power series.
18. Prove termwise integration/differentiation and uniqueness of Taylor coefficients.

One-page dependency map

Concept	Structural meaning
Metric	Gives balls, topology, and metric uniformity.
Topology	Gives neighborhoods, closure, continuity, compactness, connectedness.
Neighborhood filter	Encodes "near a point"; $\mathcal{F} \to x \iff \mathcal{N}_X(x) \subset \mathcal{F}$.
Image filter	Encodes behavior of $f(x)$ along a filter: $f_*\mathcal{F}$.
Continuity	$f_*\mathcal{N}_X(x) \to f(x)$, equivalently all convergent filters are preserved.
Ultrafilter	Maximal filter; decides every subset.
Compactness	Every ultrafilter converges; equivalently closed families with FIP intersect.
Uniformity	Filter of neighborhoods of the diagonal in $X \times X$.
Uniform continuity	Pulls entourages back to entourages.
Cauchy filter	Eventually contained in sets of arbitrarily small diameter.
Completeness	Every Cauchy filter converges.
Uniform convergence	Convergence in a function-space uniformity, often the sup metric.

Integral theorem	Continuity of a bounded linear functional.
Parameter integral	Composition $T \to C([a,b]) \to \mathbb{K}$.
Power series	Locally uniform limits of polynomials.